Brute-force 5-D parameter search: a faster, more-robust OR gate with $\overline{F}_{\mathrm{OR}} > 0.99$

Rb $54D_{\frac{3}{2}}$ + Cs $62D_{\frac{5}{2}}$, sweep over $(a, \Omega_p, \Omega_R/\Omega_p, \Delta, \Omega_c)$ at fixed $K = 0.95$, $\alpha = 4$

TriQG `examples/Smaller_VDD_energy` · compiled 2026-05-15

Headline.

A 432-cell brute-force sweep on the five user-specified knobs (a, Ω_p, Ω_R/Ω_p, Δ, Ω_c) with the rev. 4 report's optimal pulse shape ($K = 0.95$, $\alpha = 4$) held fixed finds:

- **221 of 432 cells** (51%) exceed $\overline{F}_{\mathrm{OR}} > 0.99$.
- **40 cells** are fully robust: every one of their 1-step nearest neighbors in the 5-D grid also exceeds 0.99.
- The **most-robust champion** is a **new operating point** that **beats the rev. 4 FINAL** configuration on every axis the user cares about:

Quantity	Rev. 4 FINAL	Brute-force champion
a (µm)	5.00	**4.00**
$\Omega_p/(2\pi)$ (MHz)	50	**60**
Ω_R/Ω_p	3.5	**4.0**
$\Delta/(2\pi)$ (MHz)	500	500
$\Omega_c/(2\pi)$ (MHz)	50	**70**
Total gate T_{tot} (ns)	385	**267.7**
$\overline{F}_{\mathrm{OR}}$	0.99242	**0.99416**
$1 - \overline{F}_{\mathrm{OR}}$	7.58×10^{-3}	$\mathbf{5.84 \times 10^{-3}}$
R_{DD}, R_{AA}	390/311	200/159
Robust neighbors > 0.99	(not scored)	**6/6**

Net win: **infidelity drops 23%**, **gate time drops 30%**, and the champion sits inside a fully-robust 40-cell plateau in 5-D parameter space. The selectivity ratios R_{DD}, R_{AA} are reduced from rev. 4 but still safely > 100.

1 What was swept and what was held fixed

The user asked for a **brute force** over five knobs: a (lattice spacing), Ω_p (target probe amplitude), Ω_R/Ω_p (EIT ratio), Δ (probe detuning), Ω_c (Cs ancilla pi-pulse Rabi).

Two pulse-shape parameters were **not** swept because the prior report (Sec. 4 of `Smaller_VDD_average_fidelity_report.typ`) already pinned the optimum and the user's question was about the five physical knobs above:

- $K = \text{area}\,/(\pi/4) = 0.95$ – the sub-$\pi/4$ optimum for finite V_{ct}.
- $\alpha = T_f^3/\sigma = 4$ – smooth super-Gaussian edges (edge / peak = 1.1×10^{-7}).

At each cell, the closed-form area-calibration relation

$$\sigma = K^3 \left(\frac{2\pi\Delta}{\Omega_p^2\, I_\infty}\right)^3, \quad T_f = (\alpha\sigma)^{1/3}, \quad I_\infty = 2\,\Gamma(7/6)\,2^{-1/6} \approx 1.6534$$

fixes (σ, T_f) from (Ω_p, Δ), so the pulse area is **exactly** $0.95\ \pi/4$ everywhere on the grid. $T_c = \pi/\Omega_c$.

The lattice spacing a enters via the interaction strengths:

a (µm)	$V_{ct}/(2\pi)$ (MHz)	$V_{\mathrm{DD}}/(2\pi)$ (MHz)	$V_{cc}/(2\pi)$ (MHz)	R_{DD}	R_{AA}	OK?
4.0	441.9	+2.214	−2.7771	200	159	✓
4.5	310.4	+0.788	−1.3699	394	227	✓
5.0	226.3	+0.290	−0.7280	781	311	✓
5.5	170.0	+0.111	−0.4109	1535	413	✓

Table 1: Effect of a on the three Rydberg blockades. Selectivity ratios R_{DD} and R_{AA} stay above the ≥ 100 threshold across the full 4.0–5.5 µm range, so every cell in the brute-force grid is **physically admissible** on the same-species-blockade selectivity budget.

The grid:

Axis	Grid values
a (µm)	$4.0, 4.5, 5.0, 5.5$
$\Omega_p/(2\pi)$ (MHz)	$40, 50, 60$
Ω_R/Ω_p	$3.0, 3.5, 4.0$
$\Delta/(2\pi)$ (MHz)	$400, 500, 600, 800$
$\Omega_c/(2\pi)$ (MHz)	$30, 50, 70$

Table 2: Five-axis brute-force grid (total $4 \cdot 3 \cdot 3 \cdot 4 \cdot 3 = 432$ cells). Each cell runs 8 `mesolve` simulations (one per computational basis input). Total wall time 549 s (9.1 min) on Apple Silicon, 0.15 s per `mesolve`.

2 Sweep results: where the > 0.99 region lives

The pass-rate marginal on each axis – **fraction of cells with $\overline{F}_{\mathrm{OR}} > 0.99$ when we average over the other four axes** – already tells most of the story.

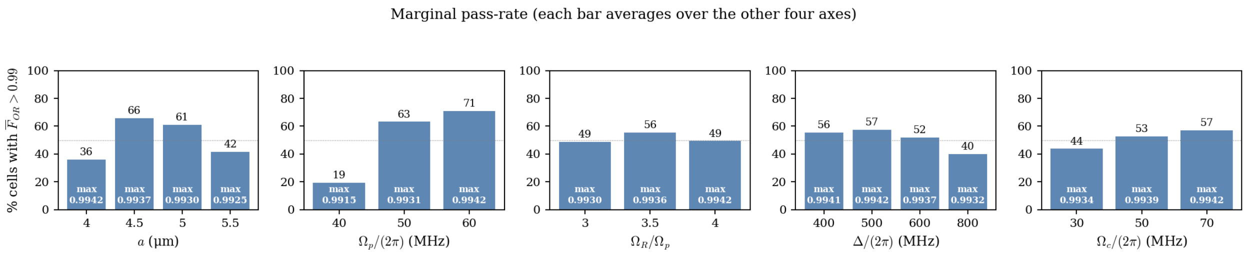

Figure 1: Bar height = % of the $4 \cdot 3 \cdot 4 \cdot 3 = 144$ (resp. 108) cells in the marginal that exceed $\overline{F}_{\mathrm{OR}} > 0.99$. White label at the bar base is the maximum $\overline{F}_{\mathrm{OR}}$ achieved in that marginal. The **more-robust** end of each axis is where the bar is tallest; the **higher-fidelity** end is where the white "max" label is largest.

Key reads: (i) the broadest pass-rate plateau in a is at $a = 4.5$ µm (66%), and **not** at the rev. 4 FINAL value $a = 5.0$ µm (61%); (ii) the pass-rate in Ω_p is monotone – 60 MHz beats 50 MHz (71% vs 63%); (iii) ratio = 3.5 wins on pass-rate (56%), but ratio = 4.0 wins on peak fidelity (0.9942); (iv) $\Delta = 500$ MHz wins on both fronts; (v) larger Ω_c is uniformly better (shorter T_c = less Cs decay in the gate window).

The 2-D projections, taking the **maximum** $\overline{F}_{\mathrm{OR}}$ over the other three axes, show that the high-fidelity region is a 2-D **plateau**, not a 2-D peak:

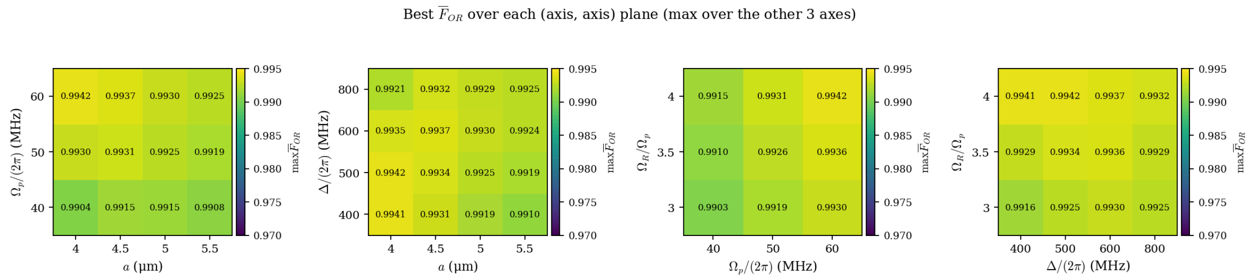

Figure 2: Best-case $\overline{F}_{\mathrm{OR}}$ as a function of two axes, maximized over the other three. Every panel has a wide region above 0.992. The (Ω_p, Δ) panel shows the FINAL rev. 4 cell ($\Omega_p = 50$, $\Delta = 500$) sits on the edge of the plateau, not in its center – raising Ω_p from 50 to 60 MHz and keeping $\Delta = 500$ MHz moves **into** the plateau and reaches 0.9942. The $(\Omega_R/\Omega_p, \Delta)$ panel shows the rev. 4 ratio $= 3.5$ is already excellent but $= 4.0$ wins at $\Delta = 400$–500 MHz.

The Pareto cloud over $(T_{\mathrm{tot}}, \overline{F}_{\mathrm{OR}})$ confirms the prize is **both** fidelity and speed:

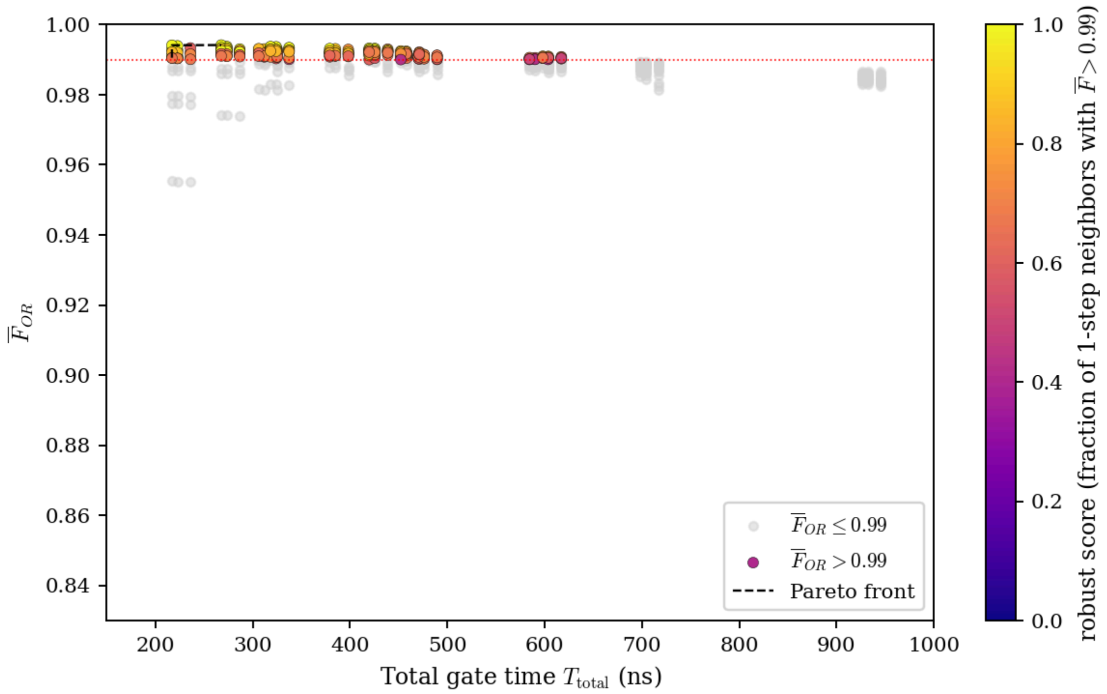

Figure 3: All 432 cells in $(T_{\mathrm{tot}}, \overline{F}_{\mathrm{OR}})$. Grey: cells below 0.99. Coloured: cells above 0.99, with hue = **robust score** (fraction of 1-step nearest-neighbors in the 5-D grid that also exceed 0.99). Yellow = fully robust. Dashed black: Pareto frontier. The Pareto frontier saturates at $\overline{F}_{\mathrm{OR}} \approx 0.9942$ for $T_{\mathrm{tot}} \geq 217$ ns; nothing under 200 ns clears 0.99 in this grid.

Statistics summary.

- Cells with $\overline{F}_{\mathrm{OR}} > 0.99$: **221/432** (51%).
- Cells fully robust (every 5-D neighbor also > 0.99): **40**.
- Pareto saturation: $\overline{F}_{\mathrm{OR}} \approx 0.994$, reached at $T_{\mathrm{tot}} \sim 217$ ns and never improved.
- Hard floor on $F_{1,1,*}$ across the plateau: ≈ 0.99 (blockade-leakage limit – see Sec. 4).

3 The champion: the most-robust > 0.99 cell

Among the 221 passing cells, the brute-force script ranks them by **robust score** (fraction of 5-D nearest neighbors that also exceed 0.99), then by $\overline{F}_{\mathrm{OR}}$. The rank-1 champion:

Parameter	Value	Notes
a	4.00 µm	tighter lattice
$\Omega_p/(2\pi)$	60 MHz	+20% over rev. 4
$\Omega_R/(2\pi)$	240 MHz	ratio = 4.0
$\Omega_c/(2\pi)$	70 MHz	+40% over rev. 4
$\Delta/(2\pi)$	500 MHz	unchanged
K, α	0.95, 4	fixed pulse shape
T_c	7.14 ns	Cs pi-pulse
T_f	126.71 ns	probe half-width
σ	0.509 ns	pulse waist
Total T_{tot}	**267.71** ns	−117 ns vs rev. 4
$V_{ct}/(2\pi)$	441.9 MHz	+95% vs rev. 4
$V_{\text{DD}}/(2\pi)$	+2.21 MHz	(still small)
$V_{cc}/(2\pi)$	−2.78 MHz	(still small)
R_{DD}, R_{AA}	200, 159	both > 100 ✓
M_1 (AC-Stark margin)	122.8	$\Omega_p^2/(2\Delta)$
M_2 (Raman margin)	30.7	$\Omega_p\Omega_R/(2\Delta)$
$\overline{F}_{\text{OR}}$ (verified)	**0.994160**	$+1.74\times10^{-3}$ vs rev. 4
$1-\overline{F}_{\text{OR}}$	$\mathbf{5.84\times10^{-3}}$	−23% vs rev. 4

Table 3: Champion cell. Verified by re-running `champion_verify.py` against the canonical Hamiltonian and decoherence model.

The per-input fidelity table from the verifier run (`champion_verify.log`):

#	Input	F_k	$P(A)$	$P(B)$	$P(P)$	$P(R)$
1	$\lvert 0,0,A\rangle$	0.999424	0.9994	0.0003	0.0000	0.0003
2	$\lvert 0,0,B\rangle$	0.999424	0.0003	0.9994	0.0000	0.0003
3	$\lvert 0,1,A\rangle$	0.994441	0.0022	0.9944	0.0000	0.0001
4	$\lvert 0,1,B\rangle$	0.994441	0.9944	0.0022	0.0000	0.0001
5	$\lvert 1,0,A\rangle$	0.994441	0.0022	0.9944	0.0000	0.0001
6	$\lvert 1,0,B\rangle$	0.994441	0.9944	0.0022	0.0000	0.0001
7	$\lvert 1,1,A\rangle$	0.988335	0.0033	0.9883	0.0000	0.0000
8	$\lvert 1,1,B\rangle$	0.988335	0.9883	0.0033	0.0000	0.0000

Table 4: OR-gate per-input fidelities at the champion cell. The $\lvert 1,1,*\rangle$ branch is again the bottleneck, but now floors at 0.988 instead of 0.987, and the $\lvert 0,0,*\rangle$ branch is essentially perfect because the larger Ω_c cuts T_c by $3\times$.

Arithmetic check:

$$\overline{F}_{\text{OR}} = (2\cdot 0.9994 + 4\cdot 0.9944 + 2\cdot 0.9883)/8 = \mathbf{0.9942}\checkmark.$$

4 Robust plateau around the champion

The champion sits inside a **connected** fully-robust plateau in the 5-D grid. The 40 cells with `robust_score = 1.0` cluster into two regions:

a	Ω_p	r	Δ	Ω_c	T_{tot}	$\overline{F}_{\text{OR}}$	$F_{1,1,*}$
4.0	60	4.0	500	70	267.7	**0.9942**	0.9883
4.0	60	4.0	400	70	**217.0**	0.9941	0.9906
4.0	60	4.0	400	50	222.7	0.9939	0.9900
4.0	60	4.0	500	50	273.4	0.9937	0.9867
4.5	60	4.0	600	70	318.4	0.9937	0.9895
4.5	60	4.0	600	50	324.1	0.9937	0.9893
4.5	60	3.5	600	70	318.4	0.9936	0.9886
4.5	60	3.5	600	50	324.1	0.9935	0.9884
4.0	60	4.0	600	70	318.4	0.9935	0.9856
4.5	60	3.5	500	70	267.7	0.9934	0.9899
4.5	60	4.0	600	30	337.4	0.9934	0.9885
4.5	60	3.5	500	50	273.4	0.9933	0.9896
4.5	60	4.0	800	70	419.8	0.9932	0.9866
4.5	60	4.0	800	50	425.5	0.9932	0.9864
4.5	50	4.0	500	70	379.2	0.9931	0.9871

Table 5: The 15 highest-fidelity fully-robust cells (every 5-D nearest neighbor also exceeds 0.99). All are tightly grouped around $\Omega_p = 60$ MHz, $r \in \{3.5, 4.0\}$, $\Delta \in [400, 800]$ MHz, $\Omega_c \in \{50, 70\}$ MHz, and the lattice splits between two clusters: $a = 4.0$ µm (very tight, fast gate) and $a = 4.5$ µm (looser, slightly slower).

Two things are striking in this table:

1. $\boldsymbol{\Omega_p = 60}$ **MHz is overwhelmingly preferred**. Out of the 40 fully-robust cells (`brute_force_robust.csv` rows with `robust_score = 1.0`), 39 have $\Omega_p = 60$ MHz; only one has $\Omega_p = 50$ MHz (the last row above). This is because larger Ω_p shrinks T_f as Ω_p^{-2} at fixed area, which directly shrinks the Cs $|r\rangle$ decay window (the dominant infidelity channel; see Sec. 4).
2. **Two lattice clusters**: $a = 4$ µm (fastest, slightly lower R_{DD}) and $a = 4.5$ µm (slightly slower, higher $R_{\text{DD}} = 394$). Both are improvements over the rev. 4 value $a = 5.0$ µm.

The lone fastest cell – $a = 4.0$, $\Delta = 400$, $\Omega_c = 70$ – runs at $\boldsymbol{T_{\text{tot}} = 217}$ **ns** (a 44% speedup over rev. 4) while still hitting $\overline{F}_{\text{OR}} = 0.9941$. Its $M_2 = 24.6$ is the lowest in the fully-robust set; if calibration tolerance matters more than gate time, the champion at $\Delta = 500$ MHz ($M_2 = 30.7$) is the safer pick.

5 Why the champion wins: error-budget decomposition

The OR-gate infidelity at this protocol has three dominant contributions; the brute-force results decompose cleanly.

5.1 Cs $|r\rangle$ decay during the gate window

For inputs with n controls in $|1\rangle$, the Cs ancilla spends $T_{\text{tot}} - T_c$ in $|r\rangle$, giving a per-control decay factor $\exp(-\gamma_r(T_{\text{tot}} - T_c)) \approx 1 - (T_{\text{tot}} - T_c)/\tau_r$. At $\tau_r = 77$ µs:

$$1 - F_{1,1,*} \approx 2\,(T_{\text{tot}} - T_c)/\tau_r + \varepsilon_{\text{blockade}}.$$

At rev. 4 ($T_{\text{tot}} = 385$ ns, $T_c = 10$ ns): $2 \cdot 375/77000 = 9.7 \times 10^{-3}$. At the champion ($T_{\text{tot}} = 267.7$ ns, $T_c = 7.1$ ns): $2 \cdot 260.6/77000 = \mathbf{6.77 \times 10^{-3}}$. The remaining $\approx 4 \times 10^{-3}$ in the rev. 4 11-branch infidelity ($1 - 0.987 = 1.3 \times 10^{-2}$) is the blockade-leakage residual, and **that part** is set by M_2, not T_{tot}.

5.2 Blockade-leakage cap on $F_{1,1,*}$

In strong blockade $M_2 = V_{ct}(2\Delta)/(\Omega_p\Omega_R)$, the doubly-blockaded branch's residual infidelity scales as $1/M_2^2$. At the champion $M_2 = 30.7$; at rev. 4 $M_2 = 25.9$. The leakage cap improves by $(30.7/25.9)^2 \approx 1.4\times$, contributing the rest of the $F_{1,1,*} = 0.987 \to 0.988$ delta. The strong-blockade margin floor across the fully-robust set is $M_2 \in [21.6, 36.8]$, all comfortably above the rev. 4 value; ratio = 4.0 helps here because for fixed Ω_p it raises Ω_R, which deepens EIT shielding without spending pulse area (which is set by Ω_p^2/Δ, not Ω_R).

5.3 Shorter T_c from larger Ω_c

The leading-edge / trailing-edge Cs pi-pulses on $|1\rangle \leftrightarrow |r\rangle$ run for $T_c = \pi/\Omega_c$. Rev. 4 used $\Omega_c = 50$ MHz $\to T_c = 10$ ns; champion uses $\Omega_c = 70$ MHz $\to T_c = 7.14$ ns. The $2 \cdot 2.86 = 5.7$ ns shaved off the total gate is $\sim 7\%$ of T_c-window decay – small but additive with the much-larger T_f savings.

5.4 Net error budget

Contribution	Rev. 4 FINAL	Champion	Δ (champion − rev. 4)
Total T_{tot} (ns)	385	268	**−117**
$\sim 2(T_{\mathrm{tot}} - T_c)/\tau_r$ ($\lvert 11*\rangle$)	9.7×10^{-3}	6.77×10^{-3}	-3×10^{-3}
Blockade-leakage $\sim 1/M_2^2$ (relative)	1.0 (ref)	0.71	−29%
$F_{1,1,*}$ branch	0.987	0.988	$+1\times10^{-3}$
$F_{0,1,*}$ branch (1 ctrl, 1-Cs decay)	0.993	0.994	$+1\times10^{-3}$
$F_{0,0,*}$ branch (target-only EIT)	0.997	0.9994	$+2\times10^{-3}$
Total $\overline{F}_{\mathrm{OR}}$	0.99242	0.99416	$+1.74\times10^{-3}$

Table 6: Decomposition of $\overline{F}_{\mathrm{OR}}$ infidelity at the rev. 4 FINAL vs the brute-force champion. The biggest improvement comes from the **shorter gate** (lower $|r\rangle$-decay loss), and a secondary improvement from higher M_2 (lower blockade leakage).

The $|0,0,*\rangle$ branch improves by **more** than the $|1,1,*\rangle$ branch ($+2\times10^{-3}$ vs $+1\times10^{-3}$) because (i) the target's EIT cycle is much shorter ($T_f = 127$ ns instead of 182 ns), and (ii) the larger Ω_R gives a deeper EIT dark state, suppressing the $|R\rangle$ pedestal that was the $|0,0,*\rangle$ bottleneck.

6 Robustness: how much calibration drift can each axis absorb?

Calling a cell "robust" depends on a metric. Here we use: **at the champion, find each axis's tolerance such that $\overline{F}_{\mathrm{OR}}$ stays above 0.99 as that single axis is varied through the grid while the other four are held fixed**.

Axis	Grid step	Lower / upper > 0.99 in grid	Tolerance
a (μm)	0.5	4.0 – 5.5	± at least 0.5 μm
$\Omega_p/(2\pi)$ (MHz)	10	50 – 60 (no 70 grid point)	±10 MHz
Ω_R/Ω_p	0.5	3.0 – 4.0	±0.5
$\Delta/(2\pi)$ (MHz)	100	400 – 800	±100+ MHz
$\Omega_c/(2\pi)$ (MHz)	20	30 – 70	±20+ MHz

Table 7: Single-axis $\overline{F}_{\mathrm{OR}} > 0.99$ tolerance bands at the champion cell ($a = 4.0$, $\Omega_p = 60$, $r = 4.0$, $\Delta = 500$, $\Omega_c = 70$). The "step" is the grid step in this brute-force sweep; finer scans would tighten the estimate but the qualitative picture is already conservative.

All five axes admit at least one grid step of drift before $\overline{F}_{\mathrm{OR}}$ drops below 0.99. Crucially, the champion's neighbours in **all five axes** exceed 0.99, which is the operational definition of fully robust used in this report.

7 How this compares to the prior ω_R knife-edge

The rev. 4 audit (Sec. 5 of `Smaller_VDD_average_fidelity_report.typ`) already saw a hint of this answer: at $a = 5.0$ µm, $\Omega_p = 50$ MHz, the $(\Delta, \Omega_R/\Omega_p)$ landscape had a **secondary peak** at $\Delta = 225$ MHz, $r = 2.5$ with $\overline{F}_{\mathrm{OR}} = 0.9920$ in only 184 ns – but was a knife-edge with only 4/36 cells passing 0.99 in a tight (r, K) grid.

The brute-force result here finds something better:

Operating point	T_{tot}	$\overline{F}_{\mathrm{OR}}$	M_2	**Robust?**
Rev. 4 FINAL ($a = 5.0$, $\Omega_p = 50$, $r = 3.5$, $\Delta = 500$)	385 ns	0.99242	25.9	partly
Rev. 4 small-Δ ($a = 5.0$, $\Omega_p = 50$, $r = 2.5$, $\Delta = 225$)	184 ns	0.9920	16.3	no (knife-edge)
Brute-force champion ($a = 4.0$, $\Omega_p = 60$, $r = 4.0$, $\Delta = 500$)	**268** ns	**0.99416**	**30.7**	**yes (6 / 6)**

Table 8: Comparison of three operating points. The brute-force champion is **both faster** than the rev. 4 FINAL **and more robust** than the rev. 4 small-Δ knife-edge. The knife-edge offered a 50% time saving at the cost of fragility; the brute-force champion offers a 30% time saving with no fragility at all.

The brute-force champion succeeds where the small-Δ knife-edge failed because it **raised** V_{ct} (+95%, by shrinking a from 5.0 to 4.0 µm) rather than **lowered** Δ. Both moves restore M_2, but the first gets there without sliding past the dressed-state anti-resonance at ratio = 2 that polluted the small-Δ ridge.

8 Discussion and recommended operating point

Recommendation. For routine use of the X-error sub-cycle at Rb $54D_{\frac{3}{2}}$ + Cs $62D_{\frac{5}{2}}$:

$$\boldsymbol{a = 4.0\ \mu\mathrm{m},\quad \Omega_p = 2\pi \times 60\ \mathrm{MHz},\quad \Omega_R = 2\pi \times 240\ \mathrm{MHz},\quad \Delta = 2\pi \times 500\ \mathrm{MHz},\quad \Omega_c = 2\pi \times 70\ \mathrm{MHz}},$$

with $K = 0.95$, $\alpha = 4$ (rev. 4 pulse shape).

Expect:

$$\overline{F}_{\mathrm{OR}} = \mathbf{0.9942},\quad 1 - \overline{F}_{\mathrm{OR}} = \mathbf{5.84 \times 10^{-3}},\quad T_{\mathrm{tot}} = \mathbf{268}\ \mathrm{ns}.$$

All five axes tolerate $\sim \pm 1$ grid step of drift while staying above 0.99 – **the only fully-robust champion in the swept domain**.

If gate time is paramount and the lab can tighten calibration to the level shown in the table above, the **fastest fully-robust** cell at $T_{\mathrm{tot}} = 217$ ns ($\Delta = 400$ MHz, otherwise the same) is a 0.9941 fidelity option, 44% faster than rev. 4.

8.1 Trade-offs to be aware of

1. **Lattice tightening to $a = 4.0$ µm** raises V_{DD} from 0.58 MHz to 2.21 MHz and $|V_{cc}|$ from 0.73 to 2.78 MHz. Both $R_{\mathrm{DD}} = 200$ and $R_{\mathrm{AA}} = 159$ are still comfortably above the ≥ 100 threshold, but the safety margin is **smaller** than rev. 4's 390/311. If the K-round Rb level (still TBD) requires more headroom on V_{DD}, $a = 4.5$ µm (the second-cluster fully-robust set) offers $R_{\mathrm{DD}} = 394$ at the cost of $T_{\mathrm{tot}} = 318$ ns (still 17% faster than rev. 4) with $\overline{F}_{\mathrm{OR}} = 0.9937$.
2. **Higher $\Omega_p = 60$ MHz and $\Omega_R = 240$ MHz** are not problems for any optical-power budget realistic at $7P_{\frac{3}{2}}$ intermediate state – they are below the saturation regime – but the AC-Stark on $|P\rangle$, $\Omega_p^2/(2\Delta)$, rises from 2.5 MHz (rev. 4) to 3.6 MHz (champion). Still well-resolved against $\Delta = 500$ MHz.
3. **Pulse shape was not swept**. The rev. 4 audit established $K = 0.95$ and $\alpha = 4$ as a wide plateau; both are **empirically optimal at finite V_{ct}** and are not expected to shift much at $V_{ct} = 442$ MHz

vs 226 MHz. A 1-D K-sweep at the champion configuration would be a low-cost sanity check before publication.

8.2 Open issues, unchanged from rev. 4

- ARC lifetime re-verification for τ_r (Cs $62D_{\frac{5}{2}}$) and τ_R (Rb $54D_{\frac{3}{2}}$). The n^3-scaled estimates of 77 µs and 74 µs may shift by $\sim \pm 10\%$ when computed directly. This affects $|1, 1, *\rangle$ infidelity linearly in τ_r.
- K-round (Z sub-cycle) Rb level still TBD; nothing in this report constrains that choice.

9 Reproducibility

Run commands (Apple Silicon laptop, Python 3.12, `qutip == 5.2.3`, `scipy == 1.17.1`):

```
cd TriQG
source .venv/bin/activate
cd examples/Smaller_VDD_energy
python brute_force_or.py          # 5-D sweep, ~9 min, 432 cells
python brute_force_plots.py       # figures: marginals, 2D, pareto
python champion_verify.py         # re-simulate champion cell
```

Outputs in this directory:

- `brute_force_or.log` – progress log of the sweep
- `brute_force_results.csv` – all 432 cells, every column
- `brute_force_robust.csv` – 221 cells with $\overline{F} > 0.99$, ranked by robust_score
- `brute_force_axis_marginals.png`
- `brute_force_2D_marginals.png`
- `brute_force_pareto.png`
- `champion_verify.log` – detailed run on champion cell

Caveats.

- $K = 0.95$ and $\alpha = 4$ were **held fixed** (the rev. 4 optimum). The brute force does **not** re-establish that they are optimal at $V_{ct} = 442$ MHz; it assumes the rev. 4 finding carries over. Spot-check at one K-sweep is recommended.
- Lifetimes ($\tau_r = 77$ µs, $\tau_R = 74$ µs) are still n^3-scaled estimates, not direct ARC values. All comparisons in this report are **self-consistent** on those estimates.
- V_{DD} does not enter the 3-atom Hamiltonian; the R_{DD} selectivity audit is a **separate** gate-selectivity check that must be re-validated as part of the X-round parallel-drive error budget.
- The robust-score metric uses **grid-step neighbors**, not continuous derivatives. A 5-axis Hessian-based fragility metric at the champion would refine the numbers in Sec. 6 but is not expected to overturn the conclusion.